

Tribhuvan University
Institute of Science and Technology
 2075
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Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 109)
 (Introduction to Information Technology)
(NEW COURSE)

Full Marks: 60
 Pass Marks: 24
 Time: 3 hours.

*Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.*

Attempt any two questions:

(2×10=20)

1. Discuss the concept behind the fixed point number representation. What can be the fixed point representation of a signed number 8? Convert $(14.14)_{10}$ into binary and octal. (3+3+4)
2. What is switching? How can you differentiate packet switching from circuit packet switching? What are the advantages of using optical fibers? (2+4+4)
3. Why system software is needed in computers? Discuss various types of operating system. (3+7)

Attempt any eight questions:

(8×5=40)

4. What is the role of Control Unit in CPU? How analog computer differs from digital? (2+3)
5. How RISC architecture differs from CISC architecture? (5)
6. What is the purpose of cache memory? How sequential access differs from direct access? (1+4)
7. Explain the working mechanism of a keyboard. (5)
8. Why IP address is used in internet? Mention the significance of domain names in internet. (2.5+2.5)
9. Define computer network. Suppose you have a two story building having 15 computers in each of two floors. Now if you are asked to create a network of these computers, what type of network will you create? Give proper justification to your answer. (2+3)
10. What is malicious software? How virus differs from worms? (1+4)
11. Discuss the characteristics of multimedia. (5)
12. What is database system? How data can be stored using relational model. (2+3)